

Continuous Time Fourier Series (CTFS):-

The Fourier series is a mathematical tool that allows the representation of any periodic signal as the sum of harmonically related signals.

The Fourier representation of periodic signals has been extended to non-periodic signals by the fundamental period, T , to ∞ and this Fourier method of representing non-periodic signals as a function of frequency is called Fourier transform.

Fourier series can be explained by two parameters :-

- (i) Trigonometric Fourier Series
- (ii) Exponential Fourier Series

Dirichlet Condition :-

The condition under which a periodic signal can be represented by a Fourier series are known as Dirichlet conditions. These conditions are as follows :-

(a) Over any period, $x(t)$ must be absolutely integrable i.e., —

$$\int_0^T |x(t)| dt < \infty$$

(b) $x(t)$ has only a finite number of maxima and minima.

(c) $x(t)$ has only a finite number of discontinuities.

Trigonometric Fourier Series :- It is

expressed as: —

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$= a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + a_3 \cos 3\omega_0 t + \dots$$

$$+ b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + b_3 \sin 3\omega_0 t + \dots$$

where $\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$ = fundamental frequency
in rad/sec

$n\omega_0$ = Harmonic Frequency

$a_0, a_n, b_n \rightarrow$ Trigonometric Fourier series coefficients.

$$a_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$$

Exponential Fourier Series :-

It can be expressed as: —

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{-jn\omega t}$$

where $C_n \rightarrow$ Fourier coefficient

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega t} dt$$

Relationship between Fourier coefficients of Trigonometric & Exponential form :-

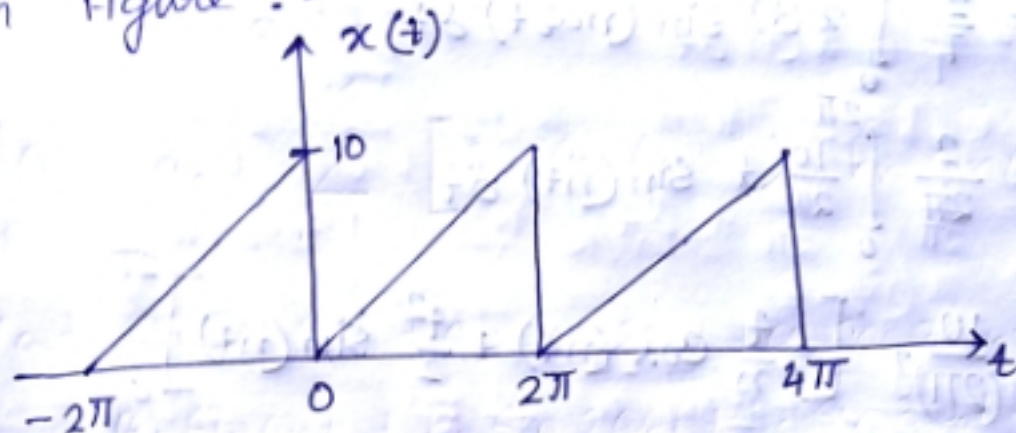
$$C_0 = a_0$$

$$C_n = \frac{1}{2} [a_n - j b_n] \text{ for } n = 1, 2, 3, 4, \dots$$

$$C_{-n} = \frac{1}{2} [a_n + j b_n] \text{ for } n = -1, -2, -3, -4, \dots$$

$$|C_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} \text{ for all } n, n \neq 0$$

Ex-1. Find the trigonometric Fourier series in figure :-



The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = \frac{2\pi}{T} = 1$

$\therefore$ The given waveform for one period may be written as —

$$x(t) = \frac{10}{2\pi} t \quad \text{for } 0 < t < 2\pi$$

The trigonometric Fourier series of $x(t)$ is :-

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$a_n = \frac{1}{T} \int_0^T x(t) \cos(n\omega_0 t) dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t \cos(nt) dt$$

$$= \left(\frac{10}{2\pi}\right)^2 \left[\frac{t}{n} \sin nt + \frac{t^2}{2} \cos nt \right]_0^{2\pi}$$

$$= \frac{10}{(2\pi)^2} [\cos(2n\pi) - \cos 0] = 0$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega t) dt$$

$$= \frac{2}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t \sin(nt) dt$$

$$= \frac{10}{(2\pi)^2} \left[-\frac{t}{n} \cos(nt) + \frac{t^2}{2} \sin(nt) \right]_0^{2\pi}$$

$$= -\frac{10}{n\pi}$$

$$x(t) =$$

$$a_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t dt = \frac{10}{(2\pi)^2} \left[\frac{t^2}{2} \right]_0^{2\pi}$$

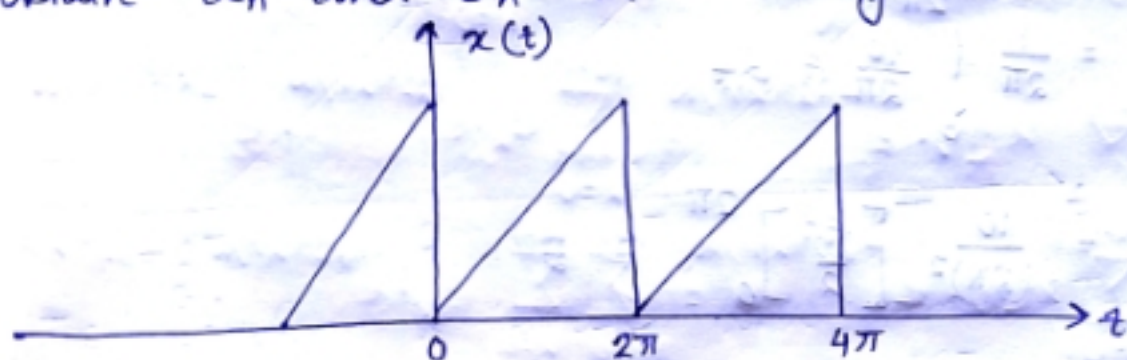
$$= 5$$

$$\therefore x(t) = 5 + \sum_{n=1}^{\infty} \left[0 - \frac{10}{n\pi} \sin(nt) \right]$$

$$x(t) = 5 - \frac{10}{\pi} \sin t - \frac{10}{2\pi} \sin 2t - \frac{10}{3\pi} \sin 3t - \dots$$

Ex-2. Find the exponential Fourier series for the waveform shown in the given figure.

Using the coefficient of the exponential series obtain a_n and b_n of the trigonometric series.



Solution:- The waveform is periodic with period $T = 2\pi$ and fundamental frequency

$$\omega_0 = \frac{2\pi}{T} = 1.$$

$\therefore$ The given waveform for one period may be written as —

$$x(t) = \frac{10}{2\pi} t \quad \text{for } 0 < t < 2\pi$$

Fourier series coefficient $C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t e^{-jnt} dt \\
 &= \left(\frac{10}{2\pi}\right)^2 \left[-t \cdot \frac{e^{-jnt}}{jn} - \frac{1}{(jn)^2} \cdot e^{-jnt} \right]_0^{2\pi} \\
 &= j \cdot \frac{10}{2n\pi}
 \end{aligned}$$

$$c_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t dt$$

$$[\because c_0 = a_0]$$

$$= \frac{10}{(2\pi)^2} \left[\frac{t^2}{2} \right]_0^{2\pi} = 5$$

$\therefore$ Exponential Fourier series for the given waveform is : —

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{\infty} c_n \cdot e^{jnt}$$

$$= \dots - j \frac{10}{4\pi} e^{-j2t} - j \frac{10}{2\pi} e^{-jt} + 5 + j \frac{10}{2\pi} e^{jt} + j \frac{10}{4\pi} e^{j2t} + \dots$$

The trigonometric Fourier series coefficients are—

$$a_0 = C_0 = 5$$

$$a_n = C_n + C_{-n} = 0$$

$$b_n = j(C_n - C_{-n}) = -\frac{10}{n\pi}$$

Discrete time Fourier series (DTFS):—

The Fourier series of discrete time periodic signal $x(n)$ with period N is defined as—

$$x(n) = \sum_{k=0}^{N-1} C_k \cdot e^{j \frac{2\pi kn}{N}} = \sum_{k=0}^{N-1} C_k \cdot e^{j\omega_0 kn}$$

$$= \sum_{k=0}^{N-1} C_k \cdot e^{j\omega_k n}$$

Where, C_k = Fourier coefficients

$\omega_0 = \frac{2\pi}{N}$ = Fundamental frequency of $x(n)$

$\omega_k = \omega_0 k = \frac{2\pi k}{N}$ = k^{th} harmonic frequency of $x(n)$

$C_k \cdot e^{j\omega_k n}$ = k^{th} harmonic component of $x(n)$

The Fourier coefficients —

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \cdot e^{-j \frac{2\pi kn}{N}} \quad \text{for } k=0,1,2,3,\dots,N-1$$

Evaluation of Fourier coefficient for discrete time signal :-

$$\text{We know that, } x(n) = \sum_{k=0}^{N-1} c_k \cdot e^{j2\pi kn/N}$$

$$\Rightarrow x(n) \cdot e^{-j2\pi kn/N} = \sum_{k=0}^{N-1} c_k \cdot e^{j2\pi kn/N} \cdot e^{-j2\pi kn/N}$$

$$\Rightarrow \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} c_k \cdot e^{j2\pi kn/N} \cdot e^{-j2\pi kn/N}$$

$$\Rightarrow \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N} = c_k \cdot N \quad [\text{when } n=k]$$

$$\therefore \boxed{c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N}}$$

Difference between CTFS and DTFS:-

- The frequency range of CTFS is from $-\infty$ to $+\infty$ i.e. it has infinite frequency spectrum.

- Whereas the frequency range of DTFS is from 0 to 2π (or $-\pi$ to $+\pi$) and so it has finite frequency spectrum.

Ex-1. Determine the Fourier series representation of the following discrete time signals.

Solⁿ:- $x(n) = 2 \sin \frac{\sqrt{3}}{2} \pi n$

$$\Rightarrow 2 \sin \frac{2\pi kn}{N} = 2 \sin \frac{\sqrt{3}}{2} \pi n$$

$$\Rightarrow \frac{k}{N} = \frac{\sqrt{3}}{4}, \text{ which is not rational}$$

$\therefore x(n)$ is non-periodic signal and so Fourier series does not exist.

$$x(n) = 3 \cos \frac{\pi n}{4}$$

$$\Rightarrow 3 \cos \frac{2\pi kn}{N} = 3 \cos \frac{\pi n}{4}$$

$$\Rightarrow \frac{k}{N} = \frac{1}{8}, \text{ which is rational}$$

Fundamental period $N=8$

$$\text{Fundamental frequency } \omega_0 = \frac{2\pi}{N} = \frac{\pi}{4}$$

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N}$$

$$= \frac{1}{8} \sum_{n=0}^7 3 \cos \frac{\pi n}{4} \cdot e^{-j2\pi kn/4}$$

$$= \frac{3}{8} \sum_{n=0}^7 \cos \frac{\pi n}{4} \cdot e^{-j2\pi kn/4} \quad \left[\frac{2\pi}{N} = \frac{\pi}{4} \right]$$

$$= \frac{3}{8} \left[(\cos 0) \cdot e^0 + (\cos \frac{\pi}{4}) \cdot e^{-j2\pi k/4} + (\cos \frac{2\pi}{4}) \cdot e^{-j4\pi k/4} \right. \\ \left. + \dots + (\cos \frac{7\pi}{4}) \cdot e^{-j14\pi k/4} \right]$$

Introduction to Fourier Transform

→ Fourier transform is a mathematical tool used for frequency analysis of signals.

→ Fourier transform is also known as the frequency domain representation of the signal.

→ Fourier transform can exist for -

- Energy signals

- Power signals

- Impulse related signals

↳ absolutely integrable signals.

↳ Not absolutely integrable signal but we can apply Fourier transform in it.

→ Fourier transform can be useful for periodic signal also.

→ Representation:-

$$X(t) \Leftrightarrow X(j\omega) \text{ or } X(f)$$

rad/sec
↓
Hz

So, the Fourier transform of a signal is a complex number so it contains a magnitude as well as angle.

$$X(j\omega) = |X(j\omega)| \angle X(j\omega)$$

Formula ① $x(t) \longrightarrow x(j\omega)$

$$x(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

↳ Fourier transform

Formula ② (Transform frequency domain signal to time domain signal)

$$x(j\omega) \longrightarrow x(t)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(j\omega) \cdot e^{j\omega t} d\omega$$

↳ Inverse Fourier transform

~~Fourier~~
~~Pair~~
Formula ① & ② — together — it is called Fourier transform pair

These are applicable for absolutely integrable signals.

② Continuous Time Fourier Transform :-

Fourier transform of any continuous signal $x(t)$ may be expressed as —

$$X(f) = F | x(t) |$$

$$= \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi ft} dt$$

or we can write as —

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

The inverse Fourier transform in terms of cyclic frequency f is written as —

$$x(t) = F^{-1} | X(f) | = \int_{-\infty}^{\infty} X(f) \cdot e^{j2\pi ft} df$$

$$\text{or } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$